Attention to the Macroeconomy

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Motivation

- Attention is key to belief formation and decision making.
  - Attention to the economy is central for business cycles, and policy transmission.
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    (Bonaglia et al., 2025; Bordalo et al., 2025; Gennaioli et al., 2024)
- Empirical properties of attention to the economy and link to beliefs not well understood due to a lack of direct data on attention allocation.

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- We test whether these links are state dependent or not, in ways that the different theories predict.

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(4) From ToM to decisions:

- Firm managers that have inflation ToM have a stronger tendency to increase prices in response to the shock.

Related literature

- **Determinants of information acquisition and attention allocation to the economy.**

  (Adam, 2007; Bracha and Tang, 2024; Capozza et al., 2022; Coibion and Gorodnichenko, 2015; Coibion et al., 2018; Flynn and Sastry, 2024; Goldstein, 2023; Korenok et al., 2023; Mikosch et al., 2024; Pfäuti, 2024a,b; Roth et al., 2022; Song and Stern, 2024; Weber et al., 2025)

  ↪ We provide direct field evidence on the dynamics, consequences, and drivers of what HHs and firms have top of mind.

- **Role of personal experiences in belief formation and attention allocation.**

  (Afrouzi et al., 2023; Andre et al., 2022; Bordalo et al., 2024, 2023a,b, 2020; Cenzon, 2025; Conlon and Patel, 2025; D'Acunto and Weber, 2022; Enke et al., 2024; Gennaioli et al., 2024; Gödker et al., 2021; Goldfayn-Frank and Wohlfart, 2020; Graeber et al., 2024; Kahana, 2012; Laudenbach et al., 2024; Malmendier and Nagel, 2016; Salle et al., 2023)

  ↪ Our results point to a state-dependent role for experiences in what people have top of mind and their belief formation.

Outline

Setting and Data

Context and timeline

- We conduct **quarterly panel surveys** with German households and firms between December 2020 and December 2024 (17 waves).
  - Firms: ifo Business Survey ($N \approx 3,000 - 3,700$ per wave).
  - Households: Dynata, broadly representative ($N \approx 1,400 - 5,700$ per wave).

  ▸ Summary statistics ▸ Responses/Attrition

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  ▸ Summary statistics ▸ Responses/Attrition

- Macroeconomic environment: ▸ More
  - Recovery from the pandemic recession.
  - Surge in inflation (largely unexpected) starting mid-2021 to $\approx$ 10% end-2022, amplified by Russia's invasion of Ukraine and energy crisis, then close to target in 2024.
  - Interest rate at the ZLB until mid-2022, then sharp rate hikes up until 4.5% in September 2023, followed by steady decline to 3.15% at end of 2024.
  - The unemployment rate slightly but steadily increased from 5% in 2022 to 6% in 2024.

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  - Information acquisition.
    (Mankiw and Reis, 2002; Reis, 2006a,b)
  - Information processing.
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- We rely on an open-ended elicitation:
  - Increasingly common in economics.
    (Andre et al., 2022, 2025, 2024; Haaland et al., 2025; Stantcheva, 2021, 2023)
  - Core advantage compared to more structured question formats: no priming through displayed response options.

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Q: ***What topics come to mind when you think about the economic situation of your company/household?***

Coding of open-ended responses

Coding scheme for responses to the open-ended question to quantitatively analyze the unstructured text data:

- Codes for a range of topics:
  - **Macroeconomic**: pandemic, inflation (incl. energy), growth, monetary policy, ...
  - **Household-level**: income, spending, saving, housing costs, ...
  - **Firm-level**: supply chain, input factors, product demand, costs, ...
- Each response can be assigned multiple codes.
- Instruct research assistants to apply the coding scheme to the open-text responses.

Reliability of coding scheme & validity of open-ended data

- 87.5% of HH responses and 98.0% of firm responses can be assigned at least one code.
- Double-coding of subset: High inter-rater reliability ($\approx$ 90%).
- Large overlap with automated word counts. ▸ More
- Large overlap with AI-based classification. ▸ More

→ **Coding scheme reliably captures content of open-ended data.**

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- Strongly correlated w/ attention measure based on structured survey question. More
- Time variation closely aligned with Google searches. More

$\rightarrow$ **Supports validity of open-ended measurement.**

Conceptual Framework

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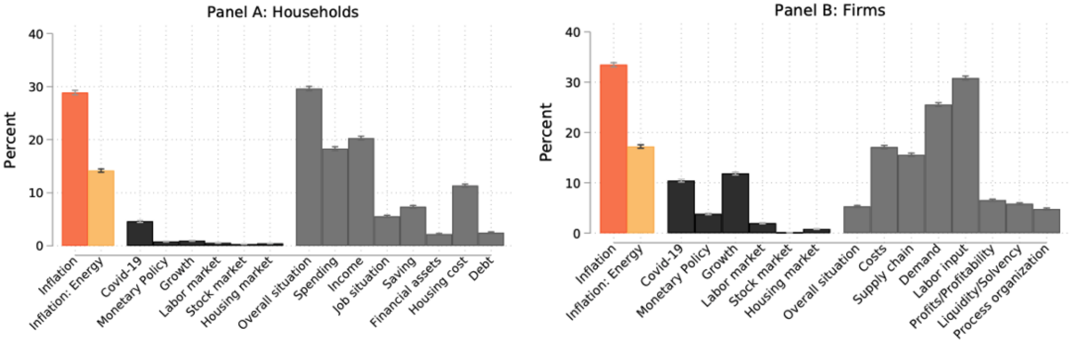
	Goal-optimal attention
Determinants of what is ToM	Payoff-relevance interacting with the volatility/uncertainty of the variable of interest
ToM and information acquisition and processing	A topic being top of mind optimally increases acquisition (or processing) of information, which is then optimally processed
ToM and beliefs	A variable being top of mind increases belief accuracy

Conceptual Framework

	Goal-optimal attention	**Non-goal-optimal attention**
Determinants of what is ToM	Payoff-relevance interacting with the volatility/uncertainty of the variable of interest	Payoff-relevance but also other determinants such as prior experiences (interacting with similarity of the context)
ToM and information acquisition and processing	A topic being top of mind optimally increases acquisition (or processing) of information, which is then optimally processed	A topic being top of mind increases information acquisition but not necessarily optimally; information not necessarily optimally processed
ToM and beliefs	A variable being top of mind increases belief accuracy	A variable being top of mind does not necessarily increase belief accuracy

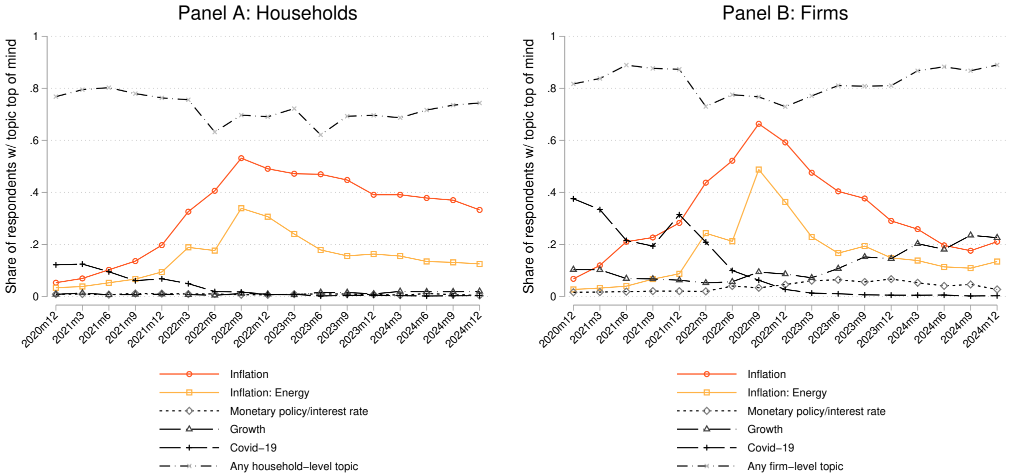
Determinants of What is Top of Mind

What is top of mind across topics



- Macro topics more likely to be ToM for firms than households.
- Macroeconomic topics dominated by inflation. Variance Decomposition Joint variation across topics

What is top of mind over time



- ToM dynamics mirror macroeconomic changes.

Determinants of what is top of mind

(1) Measures of payoff relevance:

- Many variables have ambiguous effects on payoff relevance.
- What we use: exposure to energy cost fluctuations:
  - Firms: ratio of energy costs to revenues in 2021.
  - Households: primary heating source in December 2021 was fossil (gas, oil, wood).

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(2) Measures of experiences:

- Firm managers and households: experienced the 1970s (were at least a teenager at the end of the 1970s).
- Households: say in 2021 that they have personally experienced substantial real income or wealth losses due to inflation in the past.

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Specifications:

- For each topic, we regress a ToM dummy on (1) and (2) plus controls.
  - Firm controls: number of employees (in logs), export share, dummy for firm owner, federal state.
    Household controls: gender, employment status, household income (in logs), homeowner and stockowner dummies, education, federal state.

Determinants of what is top of mind

	Households				Firms	
	(1)	(2)	(3)	(4)	(5)	(6)
ToM:	Inflation	Energy	Inflation	Growth	Inflation	Energy
Oil crises experience	0.065***	0.056***			0.006	0.015**
	(0.008)	(0.006)			(0.008)	(0.007)
High energy cost share					0.060***	0.091***
					(0.008)	(0.007)
Fossil heating	0.076***	0.053***	0.080***	0.003		
	(0.018)	(0.010)	(0.021)	(0.003)		
Any loss due to inflation			0.057***			
			(0.008)			
Income loss due to recession				0.005***		
				(0.002)		
Distinct respondents	10,760	10,760	5,755	5,737	5,057	5,057
Observations	43,354	43,354	30,470	30,380	41,663	41,663
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

- Both payoff relevance and experiences shape what HHs and firms have ToM.

Context-dependent determinants

What is top of mind depends on the context, both in theories of goal-optimal and non-goal-optimal attention.

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Subperiods:

(1) Pre-shock (Dec 2020 - June 2021): inflation close to 2%, stable energy prices.

(2) Inflation take-off (Sep - Dec 2021): Inflation increases to $> 5\%$, energy prices increase.

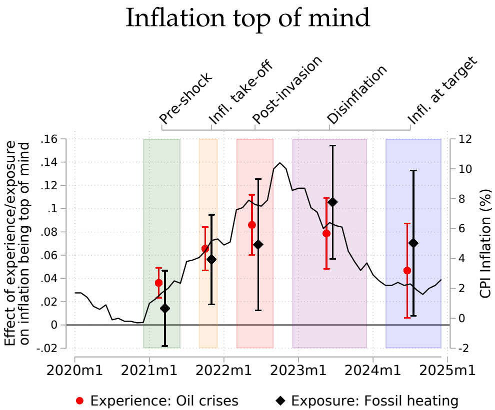
(3) Post-invasion (March - Sep 2022): Inflation $\uparrow$ 10%, skyrocketing energy prices.

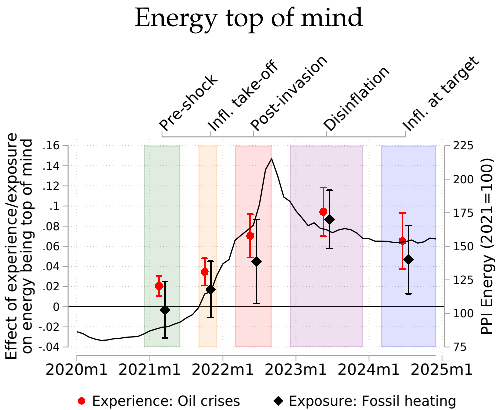
(4) Disinflation (Dec 2022 - Sep 2023): inflation high but decreasing to 4%, partial reversal of energy prices.

(5) Inflation at target (Dec 2023 - Dec 2024): inflation close to 2%, stable but elevated energy prices.

$\Rightarrow$ For each subperiod, regress ToM dummy on payoff-relevance & experience measures.

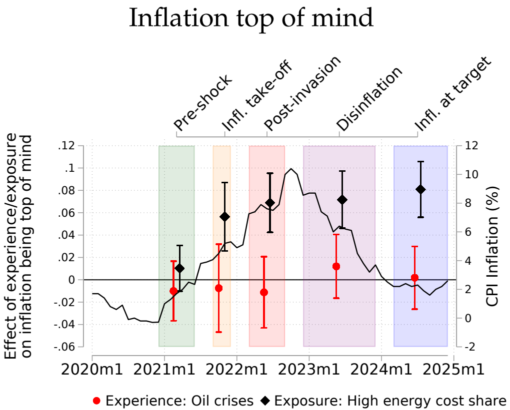
Context dependence for households

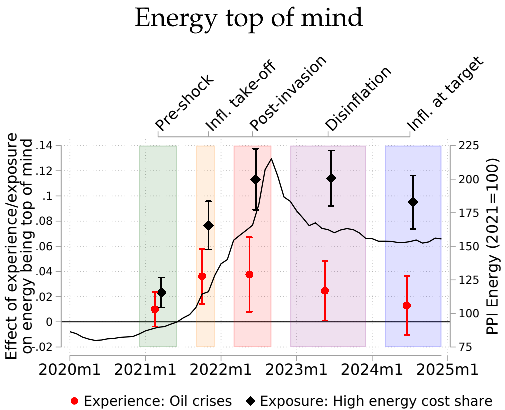




⇒ both measures become more important as inflation and energy prices increase.

Context dependence for firms





⇒ strong state-dependency of payoff-relevance measure, less for experience effects.

Consequences of What is Top of Mind

Information acquisition

- Measure of information acquisition: how often survey respondents informed themselves about inflation in last three months.
- Available: December 2020 - March 2023
- Alternative measures:
  - heard news about inflation in last three months.
  - minutes spent consuming news about inflation in previous week.
- All measures are strongly correlated. More

Information acquisition: empirical specifications

(1) Pre-shock correlations: Regress info acquisition measure on inflation ToM

+ wave fixed effects and rich baseline set of controls

$\Rightarrow$ information acquisition measures are 0.22-0.33 standard deviations higher when inflation is ToM ($p < 0.01$)

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(2) Response to the shock:

$$I_{it} = \alpha_i + \gamma_t + \sum_p \beta_p \left[\text{ToM}_{it}^{\pi} \cdot \mathbf{1}(t \in p) \right] + \delta_p \left[X_{it}' \cdot \mathbf{1}(t \in p) \right] + \varepsilon_{it}$$

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(3) Addressing reverse causality: instrument contemporaneous ToM_{it}^{π} with previous-subperiod average.

→ subperiods constitute new signals: instrument plausibly orthogonal to new developments.

Information acquisition: Results

	Households: Info. acquisition (z)		Firms: Info. acquisition (z)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Inflation top of mind				
× 1(Inflation take-off)	0.204***	0.045	0.161***	0.312**
	(0.022)	(0.197)	(0.022)	(0.147)
× 1(Post invasion)	0.213***	0.123	0.076***	0.166
	(0.018)	(0.211)	(0.017)	(0.216)
× 1(Disinflation)	0.269***	0.245	0.112***	0.285
	(0.024)	(0.206)	(0.021)	(0.225)
Standard controls interacted with periods	yes	yes	yes	yes
Time FE	yes	yes	yes	yes
Firm/Individual FE	yes	yes	yes	yes
Observations	26,432	25,269	23,034	22,557
Distinct respondents	5,662	5,311	3,634	3,492

⇒ Inflation being top of mind is more strongly associated with information acquisition during the shock periods than during the pre-shock period.

Inflation expectations

Use the same empirical approach for inflation expectations as for information acquisition.

(1) Pre-shock: no strong correlation for households, positive correlation for firm managers. ▸ Confidence

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⇒ What about the dynamics during the shock periods?

- Estimate subperiod-interaction OLS specification (2) and IV specification (3) as before.

Inflation expectations: results

	Households: Expected inflation (%)		Firms: Expected inflation (%)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Inflation top of mind				
× 1(Inflation take-off)	0.140	0.743	0.170***	1.133***
	(0.117)	(1.231)	(0.062)	(0.439)
× 1(Post invasion)	0.621***	2.084	0.237***	1.628**
	(0.109)	(1.368)	(0.060)	(0.712)
× 1(Disinflation)	0.290***	1.961	0.116***	1.201*
	(0.101)	(1.316)	(0.042)	(0.713)
× 1(Inflation at target)	0.173	1.316	-0.085*	0.979
	(0.143)	(1.301)	(0.050)	(0.874)
Standard controls interacted with periods	yes	yes	yes	yes
Time FE	yes	yes	yes	yes
Firm/Individual FE	yes	yes	yes	yes
Observations	33,008	31,368	35,194	34,259
Distinct respondents	5,697	5,323	3,745	3,556

⇒ Having inflation ToM increases expectations more strongly in response to the shock.

Deviations from benchmarks

Do these higher inflation expectations reflect a shift towards better-calibrated forecasts?

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- We need *ex-ante* benchmarks: what would have been the *best possible forecast given the available information at the time*?

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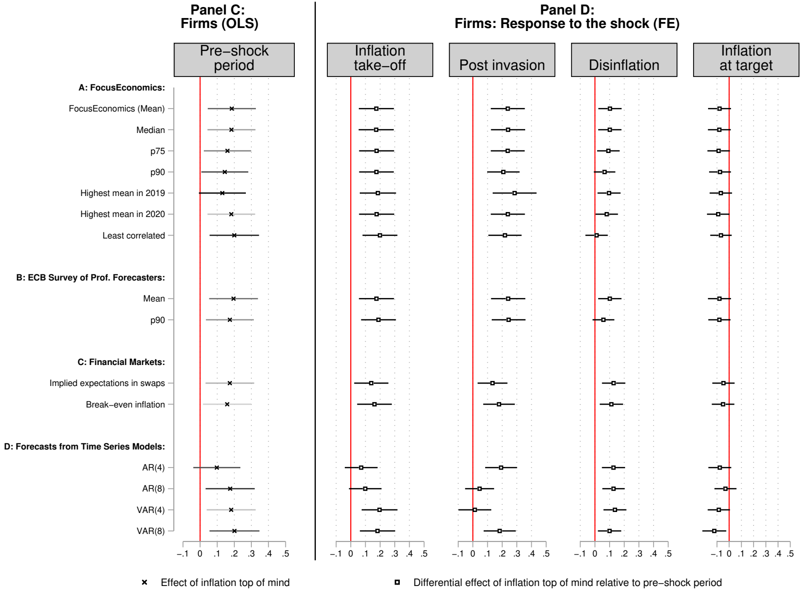
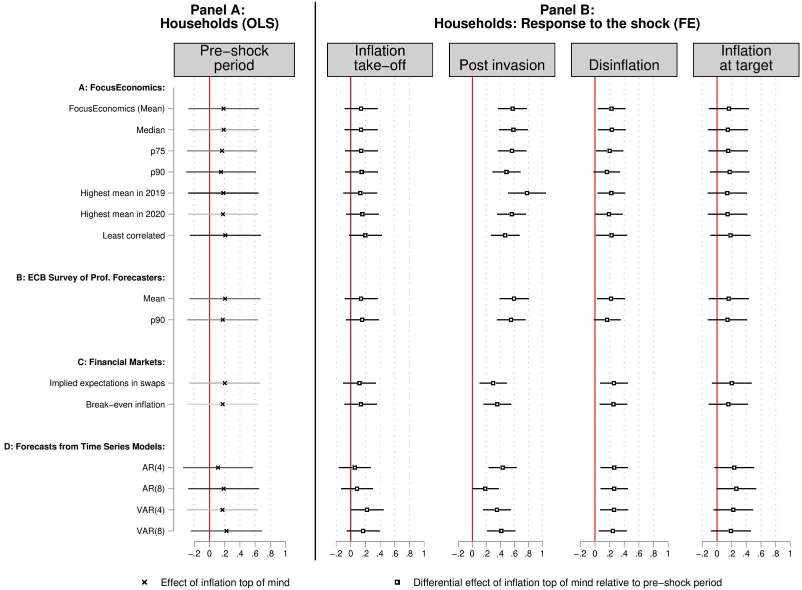
Do these higher inflation expectations reflect a shift towards better-calibrated forecasts?

- We need *ex-ante* benchmarks: what would have been the *best possible forecast given the available information at the time*?
- Consider three sets of benchmarks:
  1. professional forecasters (and subsets thereof)
  2. financial-market-implied expectations
  3. time-series models
- Focus on absolute deviations.
- Follow the same empirical strategy as before.

Deviations from benchmarks: Results for firms



Deviations from benchmarks: Results for households



How different sources of attention shape expectations

Do the sources of having inflation ToM – payoff relevance and experiences – matter for the formation of inflation expectations?

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Specification:

$$
\begin{aligned}
Y_{it} \; = \; \alpha_i \; + \; \gamma_t \; &+ \; \sum_p \beta_p \left[\mathrm{EXPOSURE}_i \cdot \mathbf{1}(t \in p) \right] \\
&+ \; \sum_p \gamma_p \left[\mathrm{EXPERIENCE}_i \cdot \mathbf{1}(t \in p) \right] \; + \; \delta_p \left[X'_{it} \cdot \mathbf{1}(t \in p) \right] \; + \; \varepsilon_{it}.
\end{aligned}
$$

- Y_{it} : inflation expectations or deviations from benchmarks.

Sources of attention and inflation expectations: Firms

	(1) $Exp^{\pi}_{i,t}$	(2) Professionals	(3) Fin. markets	(4) AR(4)
Experience: Oil crises				
× 1(Inflation take-off)	0.044	0.049	0.060	0.085
	(0.084)	(0.080)	(0.078)	(0.074)
× 1(Post invasion)	0.082	0.094	0.091	0.098
	(0.108)	(0.105)	(0.091)	(0.096)
× 1(Disinflation)	0.177**	0.153*	0.165**	0.169**
	(0.086)	(0.079)	(0.080)	(0.078)
× 1(Inflation at target)	0.015	-0.003	0.000	0.036
	(0.076)	(0.069)	(0.070)	(0.069)
Controls interacted with periods	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Firm/Individual FE	Yes	Yes	Yes	Yes
Yes				
Observations	35,468	35,468	35,468	35,468
Distinct respondents	3,124	3,124	3,124	3,124

⇒ Experiences shift expectations away from benchmarks, esp. in disinflationary phase.

- Experience effects more pronounced for households. Households

Sources of attention and inflation expectations: Firms

	(1) $Exp^{\pi}_{i,t}$	(2) Professionals	(3) Fin. markets	(4) AR(4)
Actual relevance				
× 1(Inflation take-off)	0.115	0.148**	0.099	0.060
	(0.072)	(0.070)	(0.068)	(0.065)
× 1(Post invasion)	0.270***	0.315***	0.155**	0.252***
	(0.091)	(0.089)	(0.078)	(0.083)
× 1(Disinflation)	0.157**	0.192***	0.164**	0.165**
	(0.075)	(0.071)	(0.071)	(0.070)
× 1(Inflation at target)	-0.023	-0.007	-0.021	-0.031
	(0.067)	(0.063)	(0.064)	(0.062)
Controls interacted with periods	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Firm/Individual FE	Yes	Yes	Yes	Yes
Observations	35,468	35,468	35,468	35,468
Distinct respondents	3,124	3,124	3,124	3,124

⇒ Payoff relevance shifts expectations away from benchmarks.

- Similar for households. Households

Mechanisms behind non-goal-optimality

- Want to understand what people think about when forming their expectations.

Mechanisms behind non-goal-optimality

- Want to understand what people think about when forming their expectations.
- So, we asked them:

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... and are less likely to say that they just guessed.

- What about the dynamics of those patterns over the shock periods?

Dynamics of selective focus on inflationary forces

	Topic in mind when forming inflation expectations					
	(1) Energy	(2) Monetary policy	(3) Fiscal policy	(4) Geopolitics	(5) Recent inflation	(6) Guess
Inflation top of mind						
$\times$ 1(t $\in$ {2021m9–2021m12})	0.035***	0.004	0.017**	-0.015***	0.004	-0.027***
	(0.011)	(0.007)	(0.008)	(0.005)	(0.013)	(0.009)
$\times$ 1(t $\in$ {2022m3–2022m12})	0.046***	0.008**	-0.013***	0.046***	0.033***	-0.021***
	(0.008)	(0.004)	(0.004)	(0.008)	(0.008)	(0.007)
Controls interacted with periods	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Distinct respondents	5,630	5,630	5,630	5,630	5,630	5,630
Observations	24,584	24,584	24,584	24,584	24,584	24,584

From inflation being ToM to decisions

- How does having inflation ToM shape firms' price-setting behavior?
- We estimate:

$$\mathbf{1}(\mathit{PriceIncrease}_{it}) = \alpha_i + \gamma_t + \sum_p \beta_p \left[\text{ToM}^{\pi}_{it} \cdot \mathbf{1}(t \in p)\right] + \delta_p \left[X'_{it} \cdot \mathbf{1}(t \in p)\right] + \varepsilon_{it}$$

- $\mathbf{1}(\mathit{PriceIncrease}_{it})$: dummy indicating whether the firm says that they have raised the price of their main product in the last three months.

From inflation being ToM to pricing decisions: Results

	Price increase		
	(1) OLS	(2) OLS	(3) IV
Panel A: Pre-shock period			
Inflation top of mind	0.147***		
	(0.017)		
Panel B: Response to the shock			
× 1(Inflation take-off)		0.087***	0.243**
		(0.015)	(0.104)
× 1(Post invasion)		0.053***	0.375***
		(0.012)	(0.137)
× 1(Disinflation)		0.007	0.183
		(0.008)	(0.142)
× 1(Inflation at target)		0.002	0.228
		(0.011)	(0.178)
Standard controls (interacted with periods)	yes	yes	yes
Time FE	yes	yes	yes
Firm/Individual FE		yes	yes
Observations	7,351	35,301	34,358
Distinct respondents	3,505	3,717	3,529

Summary

- New panel on what HHs and firms have top of mind using open-ended questions.
- We find support for both theories of goal-optimal and non-goal-optimal attention:
  - Payoff relevance and prior experiences matter for attention allocation and belief formation in a state-dependent way.
  - Having inflation top of mind and acquiring more information about it does not necessarily improve belief accuracy...

  ... even when the underlying driver is true payoff relevance: indication of non-optimal information processing.
- **Implications:**
  - Macro modeling: Examine aggregate implications of non-goal-optimal attention.
  - Methodological: Use of open-ended data to measure economic attention in other contexts.

Thank you!

Appendix

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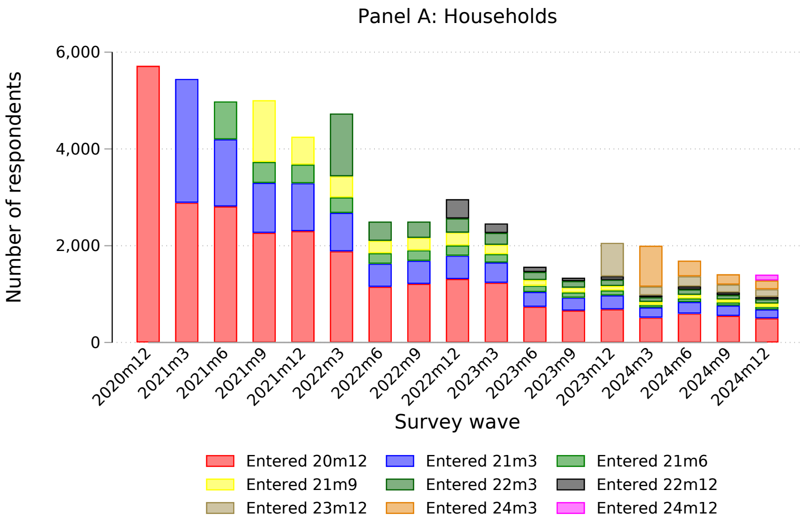
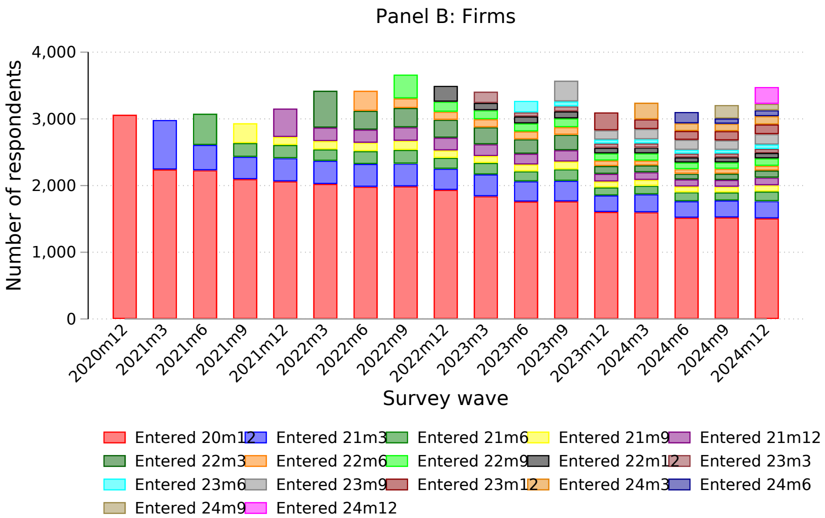
Survey responses across waves: Households

Back



Survey responses across waves: Firms

▸ Back ▸ Households



Summary Statistics

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	GSOEP	Survey samples					
	(1) Mean	(2) Mean	(3) p25	(4) Median	(5) p75	(6) SD	(7) N
Panel A: Households							
Female	0.51	0.45	0.00	0.00	1.00	0.50	51,617
Age	51.19	51.80	40.00	50.00	60.00	13.42	51,617
East	0.17	0.17	0.00	0.00	0.00	0.38	51,617
Log(HH net income)	7.96	7.84	7.60	8.01	8.36	0.67	51,617
At least highschool	0.39	0.51	0.00	1.00	1.00	0.50	51,594
Employed	0.64	0.68	0.00	1.00	1.00	0.47	49,877
Homeowner	0.49	0.49	0.00	0.00	1.00	0.50	48,930
Stockowner	0.26	0.44	0.00	0.00	1.00	0.50	48,930
Exposure: Fossil heating		0.87	1.00	1.00	1.00	0.33	19,415
Experience: Oil crises		0.45	0.00	0.00	1.00	0.50	52,008
Experience: Inflation loss		0.51	0.00	1.00	1.00	0.50	34,302
Experience: Recession loss		0.29	0.00	0.00	1.00	0.46	34,183
Panel B: Firms							
Employees		272.20	12.00	38.00	118.00	1866.27	55,268
Export share		0.17	0.00	0.04	0.25	0.25	50,671
East		0.14	0.00	0.00	0.00	0.34	50,671
Manufacturing firm		0.30	0.00	0.00	1.00	0.46	55,580
Services firm		0.39	0.00	0.00	1.00	0.49	55,580
Construction firm		0.09	0.00	0.00	0.00	0.29	55,580
Retail/wholesale firm		0.22	0.00	0.00	0.00	0.41	55,580
Respondent is firm owner		0.51	0.00	1.00	1.00	0.50	45,561
Exposure: Energy cost share (pre-shock)		0.06	0.01	0.04	0.08	0.08	37,424
Manager's experience: Oil crises		0.71	0.00	1.00	1.00	0.45	35,554

Unexpected shock to inflation

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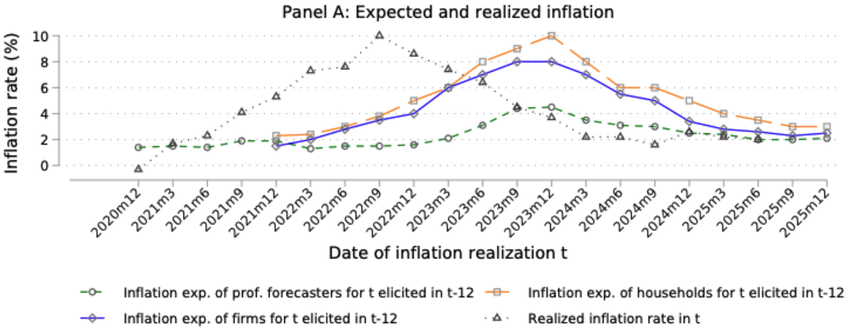
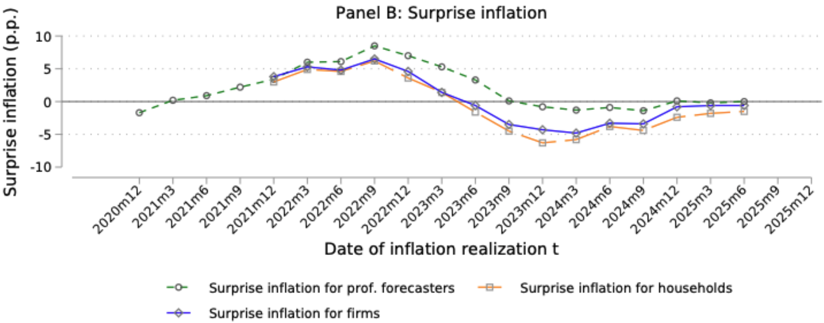


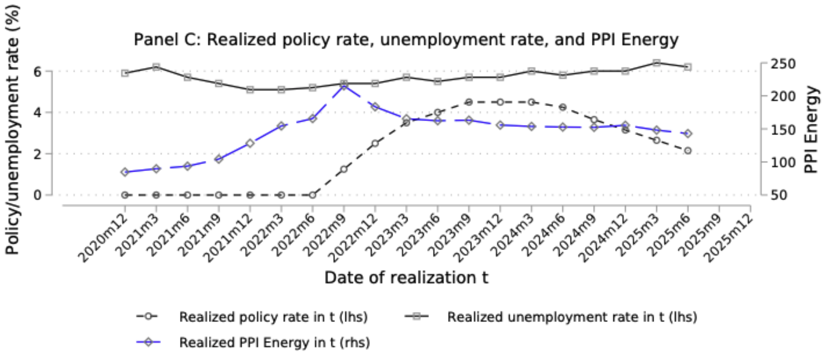
Unexpected shock to inflation

Back



Macroeconomic environment

Back



Hand-coding vs. word count for topic "inflation"

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	Hand-coded	Automated word count						Correlation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Inflation	Price	Cost	Energy	Expensive	Joint word count	hand-coded vs. joint word count
Panel A: Households								
Wave 1: 2020m12	0.05	0.01	0.02	0.01	0.03	0.03	0.08	0.77
Wave 2: 2021m3	0.07	0.01	0.02	0.01	0.04	0.03	0.09	0.82
Wave 3: 2021m6	0.10	0.02	0.04	0.02	0.05	0.04	0.12	0.87
Wave 4: 2021m9	0.14	0.04	0.05	0.02	0.06	0.05	0.16	0.87
Wave 5: 2021m12	0.20	0.07	0.07	0.02	0.09	0.04	0.21	0.93
Wave 6: 2022m3	0.33	0.09	0.14	0.04	0.18	0.06	0.33	0.94
Wave 7: 2022m6	0.41	0.21	0.17	0.05	0.17	0.06	0.43	0.91
Wave 8: 2022m9	0.53	0.20	0.20	0.08	0.32	0.06	0.53	0.94
Wave 9: 2022m12	0.49	0.23	0.19	0.06	0.30	0.07	0.52	0.92
Wave 10: 2023m3	0.47	0.23	0.18	0.06	0.25	0.08	0.51	0.91
Wave 11: 2023m6	0.47	0.28	0.16	0.06	0.18	0.06	0.47	0.91
Wave 12: 2023m9	0.45	0.25	0.14	0.05	0.16	0.07	0.45	0.90
Wave 13: 2023m12	0.39	0.19	0.13	0.06	0.17	0.07	0.39	0.92
Wave 14: 2024m3	0.39	0.18	0.11	0.05	0.16	0.08	0.40	0.90
Wave 15: 2024m6	0.38	0.19	0.12	0.06	0.13	0.06	0.38	0.92
Wave 16: 2024m9	0.37	0.17	0.09	0.05	0.13	0.08	0.38	0.87
Wave 17: 2024m12	0.33	0.16	0.10	0.05	0.12	0.07	0.35	0.91
Total (Waves 1-17)	0.29	0.12	0.10	0.04	0.14	0.06	0.30	0.92
Panel B: Firms								
Wave 1: 2020m12	0.07	0.01	0.04	0.01	0.03	0.03	0.12	0.70
Wave 2: 2021m3	0.12	0.01	0.07	0.01	0.04	0.04	0.17	0.78
Wave 3: 2021m6	0.21	0.02	0.15	0.03	0.04	0.03	0.25	0.87
Wave 4: 2021m9	0.23	0.03	0.14	0.04	0.06	0.06	0.30	0.80
Wave 5: 2021m12	0.28	0.07	0.16	0.04	0.08	0.02	0.32	0.91
Wave 6: 2022m3	0.44	0.09	0.24	0.07	0.21	0.02	0.47	0.88
Wave 7: 2022m6	0.52	0.19	0.24	0.07	0.18	0.03	0.55	0.89
Wave 8: 2022m9	0.66	0.19	0.28	0.10	0.45	0.02	0.69	0.90
Wave 9: 2022m12	0.59	0.20	0.22	0.09	0.33	0.02	0.59	0.91
Wave 10: 2023m3	0.48	0.20	0.16	0.06	0.21	0.02	0.50	0.90
Wave 11: 2023m6	0.40	0.15	0.13	0.07	0.16	0.02	0.44	0.86
Wave 12: 2023m9	0.38	0.15	0.14	0.08	0.19	0.04	0.45	0.82
Wave 13: 2023m12	0.29	0.08	0.11	0.08	0.15	0.04	0.37	0.79
Wave 14: 2024m3	0.26	0.07	0.10	0.08	0.14	0.04	0.35	0.74
Wave 15: 2024m6	0.20	0.05	0.09	0.06	0.11	0.03	0.29	0.73
Wave 16: 2024m9	0.18	0.04	0.08	0.05	0.12	0.04	0.28	0.71
Wave 17: 2024m12	0.21	0.04	0.10	0.07	0.14	0.04	0.32	0.71
Total (Waves 1-17)	0.34	0.10	0.15	0.06	0.16	0.03	0.39	0.84

Hand-coding vs. AI-based coding

Back

	AI-coded				
	(1) Covid-19	(2) Inflation	(3) Growth	(4) Any macro topic	(5) Any house-hold-level topic
AI-coded: Covid-19	0.997***	-0.079	-0.004		
	(0.004)	(0.070)	(0.007)		
AI-coded: Inflation	-0.006	0.808***	0.015		
	(0.006)	(0.032)	(0.013)		
AI-coded: Growth	-0.003	0.421**	0.746***		
	(0.004)	(0.205)	(0.219)		
AI-coded: Any macro topic				0.657***	-0.080*
				(0.057)	(0.044)
AI-coded: Any household-level topic				0.095*	0.767***
				(0.054)	(0.059)
Observations	200	200	200	200	200
R-squared	0.66	0.52	0.75	0.38	0.65

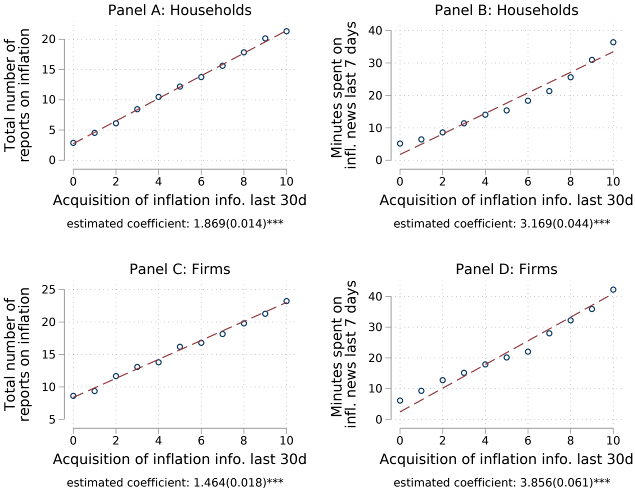
Validation: Structured question on attention

Back

	Open-ended					
	(1) Covid-19	(2) Inflation	(3) Monetary policy	(4) Growth	(5) Any macro topic	(6) Any household-level topic
Structured: Covid-19	0.098*	-0.032	-0.012*	0.012		
	(0.053)	(0.086)	(0.007)	(0.040)		
Structured: Inflation	0.008*	0.159***	0.008*	0.002		
	(0.005)	(0.041)	(0.004)	(0.014)		
Structured: Monetary policy	-0.008	0.040	0.032	0.039*		
	(0.005)	(0.059)	(0.024)	(0.023)		
Structured: Growth	-0.018*	0.089	-0.006	0.072**		
	(0.010)	(0.062)	(0.020)	(0.029)		
Structured: Any macro topic					0.151***	-0.032
					(0.049)	(0.050)
Structured: Any household-level topic					-0.072	0.469**
					(0.203)	(0.192)
Observations	468	468	468	468	468	468
R-squared	0.10	0.04	0.02	0.04	0.01	0.02
Mean dep. var.	0.01	0.26	0.01	0.03	0.29	0.79

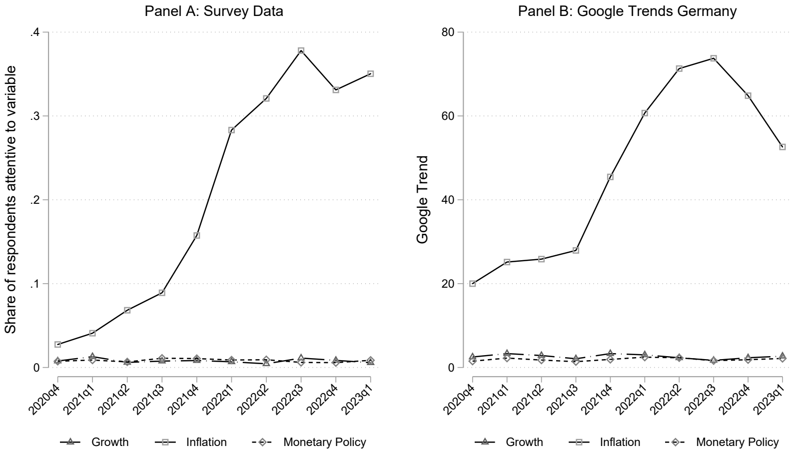
Validation: Information acquisition and news consumption

Back



Validation: Attention and Google searches

Back



How different sources shape expectations (any loss experience) ▸ Back

	Households			
	(1)	(2)	(3)	(4)
	Expected inflation next 12 months	Abs. dev. from consensus forecast (Focus)	Abs. dev. from swap-based market expect-ations	Abs. dev. from AR(4) forecast
Experience: Inflation loss				
× 1(Inflation take-off)	0.200	0.212	0.139	0.098
	(0.135)	(0.132)	(0.131)	(0.129)
× 1(Post invasion)	0.572***	0.544***	0.337**	0.465***
	(0.174)	(0.171)	(0.164)	(0.165)
× 1(Disinflation)	0.369*	0.317	0.326*	0.372*
	(0.201)	(0.195)	(0.195)	(0.194)
× 1(Inflation at target)	0.008	-0.017	-0.000	0.042
	(0.242)	(0.235)	(0.234)	(0.233)
Exposure: Fossil heating				
× 1(Inflation take-off)	0.683*	0.708*	0.700*	0.682*
	(0.402)	(0.391)	(0.387)	(0.380)
× 1(Post invasion)	1.008**	1.006**	0.972**	0.917**
	(0.484)	(0.471)	(0.458)	(0.458)
× 1(Disinflation)	0.865*	0.944**	0.934**	0.853*
	(0.461)	(0.440)	(0.443)	(0.442)
× 1(Inflation at target)	0.614	0.574	0.555	0.508
	(0.541)	(0.512)	(0.516)	(0.514)
Standard controls interacted with periods	Yes	Yes	Yes	Yes
Age interacted with periods	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
Observations	33,723	33,723	33,723	33,723
Distinct respondents	5,405	5,405	5,405	5,405

How different sources shape expectations (oil crises) ▸ Back

	Households				Firms			
	(1) Expected inflation next 12 months	(2) Abs. dev. from consensus forecast (Focus)	(3) Abs. dev. from swap-based market expect-ations	(4) Abs. dev. from AR(4) forecast	(5) Expected inflation next 12 months	(6) Abs. dev. from consensus forecast (Focus)	(7) Abs. dev. from swap-based market expect-ations	(8) Abs. dev. from AR(4) forecast
Experience: Oil crises								
× 1(Inflation take-off)	0.647***	0.627***	0.565***	0.482***	0.044	0.049	0.060	0.085
	(0.142)	(0.140)	(0.139)	(0.137)	(0.084)	(0.080)	(0.078)	(0.074)
× 1(Post invasion)	1.050***	0.941***	0.655***	0.754***	0.082	0.094	0.091	0.098
	(0.174)	(0.169)	(0.162)	(0.164)	(0.108)	(0.105)	(0.091)	(0.096)
× 1(Disinflation)	0.895***	0.795***	0.803***	0.809***	0.177**	0.153*	0.165**	0.169**
	(0.191)	(0.186)	(0.186)	(0.185)	(0.086)	(0.079)	(0.080)	(0.078)
× 1(Inflation at target)	0.405**	0.369*	0.397**	0.410**	0.015	-0.003	0.000	0.036
	(0.200)	(0.196)	(0.195)	(0.194)	(0.076)	(0.069)	(0.070)	(0.069)
Actual relevance								
× 1(Inflation take-off)	0.725*	0.737**	0.744**	0.710**	0.115	0.148**	0.099	0.060
	(0.378)	(0.367)	(0.363)	(0.357)	(0.072)	(0.070)	(0.068)	(0.065)
× 1(Post invasion)	1.067**	1.041**	0.968**	0.932**	0.270***	0.315***	0.155**	0.252***
	(0.467)	(0.453)	(0.438)	(0.439)	(0.091)	(0.089)	(0.078)	(0.083)
× 1(Disinflation)	0.937**	0.993**	0.995**	0.924**	0.157**	0.192***	0.164**	0.165**
	(0.441)	(0.419)	(0.423)	(0.422)	(0.075)	(0.071)	(0.071)	(0.070)
× 1(Inflation at target)	0.731	0.695	0.679	0.628	-0.023	-0.007	-0.021	-0.031
	(0.517)	(0.489)	(0.492)	(0.490)	(0.067)	(0.063)	(0.064)	(0.062)
Controls interacted with periods	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm/Individual FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	38,089	38,089	38,089	38,089	35,468	35,468	35,468	35,468
Distinct respondents	6,470	6,470	6,470	6,470	3,124	3,124	3,124	3,124

Variance decomposition of what is ToM

Back

	Households R^2				Firms R^2			
	(1) Ind. FE	(2) Time FE	(3) Time FE + Ind. FE	(4) Obs.	(5) Ind. FE	(6) Time FE	(7) Time FE + Ind. FE	(8) Obs.
Any macro topic	41.5	5.4	45.1	41,906	30.5	2.1	32.4	47,946
Inflation	40.1	11.8	47.7	41,906	29.3	12.5	40.5	47,946
Inflation: Energy	34.2	6.3	39.0	41,906	32.2	10.1	41.5	47,946
Covid-19	34.2	3.9	36.5	41,906	24.9	17.1	38.2	47,946
Monetary policy	23.8	0.1	23.9	41,906	30.0	0.8	30.6	47,946
Growth	22.4	0.2	22.5	41,906	24.3	3.3	26.9	47,946
Any HH-/firm-level topic	37.8	1.5	38.8	41,906	27.6	1.9	28.9	47,946

Joint variation across topics

Back

	Attention to inflation				Attention to monetary policy		Attention to any macro topic	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Households								
Attention to growth	0.118***	0.030			0.021**	0.019**		
	(0.024)	(0.024)			(0.008)	(0.008)		
Attention to monetary policy			0.271***	0.126***				
			(0.027)	(0.026)				
Attention to any HH-level topic							-0.181***	-0.262***
							(0.007)	(0.006)
Distinct respondents	10,760	7,890	10,760	7,890	10,760	7,890	10,760	7,890
Observations	43,354	41,906	43,354	41,906	43,354	41,906	43,354	41,906
R-squared	0.14	0.48	0.14	0.48	0.00	0.24	0.09	0.48
Panel B: Firms								
Attention to growth	0.020***	-0.014**			0.025***	0.009**		
	(0.007)	(0.007)			(0.004)	(0.004)		
Attention to monetary policy			0.180***	0.105***				
			(0.014)	(0.013)				
Attention to any firm-level topic							-0.294***	-0.272***
							(0.006)	(0.006)
Distinct respondents	7,855	6,266	7,855	6,266	7,855	6,266	7,855	6,266
Observations	49,535	47,946	49,535	47,946	49,535	47,946	49,535	47,946
R-squared	0.15	0.41	0.15	0.41	0.02	0.31	0.08	0.36
Controls	Yes	No	Yes	No	Yes	No	Yes	No
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual/Firm FE	No	Yes	No	Yes	No	Yes	No	Yes

Confidence

Back

	Households: Confidence (z)			Firms: Confidence (z)		
	(1) OLS	(2) OLS	(3) IV	(4) OLS	(5) OLS	(6) IV
Panel A: Pre-shock period						
Inflation top of mind	0.214***			0.071**		
	(0.037)			(0.036)		
Panel B: Reaction to the shock						
Inflation top of mind						
× 1(Inflation take-off)		0.020	0.002		0.075***	0.368**
		(0.023)	(0.224)		(0.028)	(0.188)
× 1(Post invasion)		0.044**	0.034		0.028	0.382
		(0.019)	(0.238)		(0.022)	(0.252)
× 1(Disinflation)		0.044**	-0.034		-0.005	0.401
		(0.019)	(0.232)		(0.017)	(0.258)
× 1(Inflation at target)		0.028	-0.107		-0.033	0.487
		(0.030)	(0.230)		(0.025)	(0.330)
Standard controls	yes			yes		
Standard controls interacted with periods		yes	yes		yes	yes
Time FE	yes	yes	yes	yes	yes	yes
Firm/Individual FE		yes	yes		yes	yes
Observations	8,669	33,008	31,368	7,206	33,743	32,851
Distinct respondents	5,323	5,697	5,323	3,488	3,709	3,525